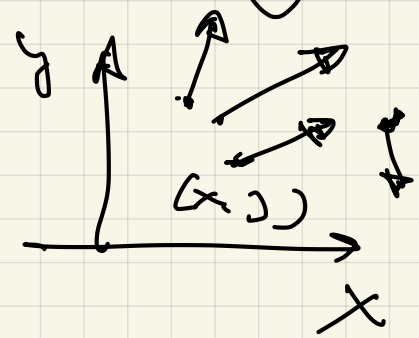
Math 534 H - Spring 2026 - Lecture 4

Variable coefficients Transport Equations (Homog.).



Suppose $a(x,y)\,u_x + b(x,y)\cdot u_y = 0$

As before $\underbrace{(a(x,y), b(x,y))}_{=\vec{V}(x,y)} \cdot \nabla_{xy} u = 0$

Now we are going to have CHARACTERISTIC CURVES

Question: How do we solve this. ?

Example: $y\,u_x + x\,u_y = 0$ $\quad a(x,y) = y$, $b(x,y) = x$

$$(y, x)\cdot \nabla_{xy} u = 0$$

$\vec{V}(x,y) = (y, x)$

$$\boxed{D_{\vec{V}(x,y)}\, u = 0} \quad (*)$$

(*) means is that $u(x,y)$ is <u>constant along curves</u> (characteristic curves) whose <u>tangent vector</u> at (x,y) is $\vec{v}(x,y) = (y, x)$ (**)

Let's find the equation of these curves:

(**) the slope of the tangent line to the curves at (x,y) is given by: $\frac{x}{y}$.

So if $\boxed{y(x)}$ represents the <u>equation of the curve</u>

then $\boxed{\underbrace{\frac{dy}{dx}}_{y'} = \frac{x}{y}} \Longrightarrow \boxed{y' \cdot y = x}$

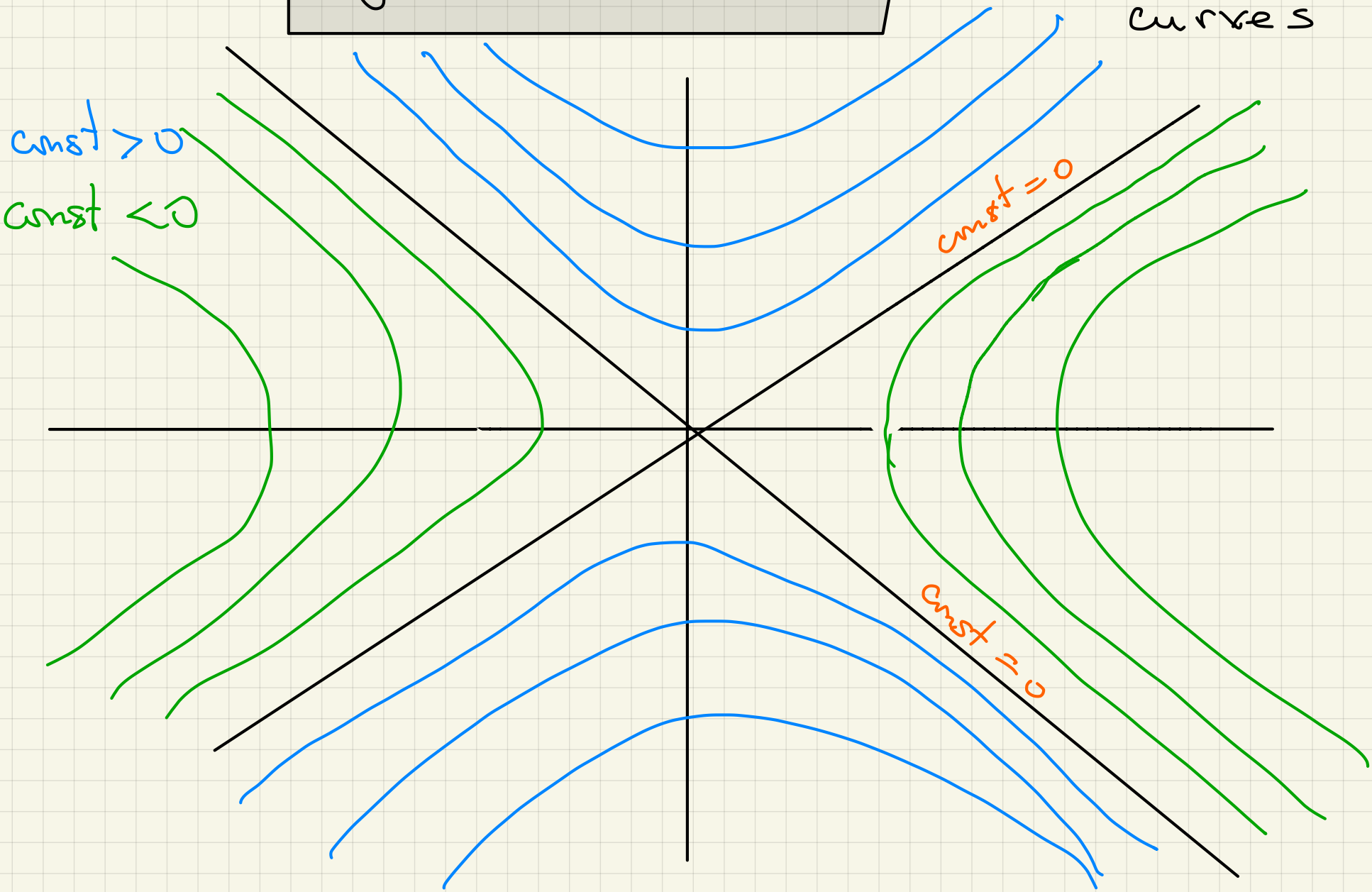
$$\left(\frac{1}{2} y^2\right)' = x$$

Integrating in $x \Longrightarrow \boxed{\frac{1}{2} y^2 = \frac{1}{2} x^2 + \text{const}}$

Hence the characteristic curves are

$$\boxed{y^2 - x^2 = \text{const}}$$

What are these curves?



Just as before then $\boxed{u(x,y) = f(y^2 - x^2)}$

where f any function of one variable.

* general solution of

$$\boxed{y u_x + x u_y = 0}$$

Check:

$u_x = f'(y^2-x^2)\cdot(-2x)$

$u_y = f'(y^2-x^2)\cdot(2y)$. So $y\cdot u_x = f'(y^2-x^2)\cdot(-2xy)$

$+ \quad x u_y = f'(y^2-x^2)\cdot(2xy)$

$= 0$.

So indeed $\boxed{y u_x + x u_y = 0}$

Now suppose that we impose that $u(0,y) = e^{-y^2}$

Then $e^{-y^2} = u(0,y) = f(y^2) \Rightarrow f(w) = e^{-w}$

$w \in \mathbb{R}$

Hence $\boxed{u(x,y) = e^{-(y^2-x^2)}}$ is the solution

to
$$\begin{cases} y\,u_x + x\,u_y = 0 \\ u(0,y) = e^{-y^2} \end{cases}$$

Back to the general case $a(x,y)\,u_x + b(x,y)\,u_y = 0$

We found that as long as we can solve

$$\boxed{\frac{dy}{dx} = \frac{b(x,y)}{a(x,y)}}$$

$\longrightarrow$ will give us the equation of the <u>Characteristic curves</u>

and thus allow us to find $u(x,y)$

<u>Example 2</u> :

$$(*)\quad \begin{cases} 2u_x + 8xy^2 u_y = 0 \\ u(0,y) = 3y \end{cases}$$

$\vec{v} = (2, 8xy^2)$

Find $u(x,y)$ solving $(*)$. To find the general solution we need to find the characteristic curves. We need to solve

$(y = y(x))$

$$\frac{dy}{dx} = \frac{8xy^2}{2} = 4xy^2$$

$$\frac{dy}{y^2} = 4x\,dx$$

$\boxed{y \neq 0}$

Domain of validity

Characteristic curves

$$-\frac{1}{y} = 2x^2 + \text{const} \qquad \boxed{2x^2 + \frac{1}{y} = \text{const}}$$

Hence $u(x,y) = f\left(2x^2 + \frac{1}{y}\right); \quad y \neq 0$

where f is any function of one variable solves

$$2u_x + 8xy^2 u_y = 0.$$

But now we want the u that also satisfies $u(0,y) = 3y$

$$3y = u(0,y) = f\left(\frac{1}{y}\right) \Rightarrow \boxed{f(w) = \frac{3}{w} \quad w \in \mathbb{R}}$$

Hence the solution $u = u(x,y)$ that satisfies (*) (both the equation and the initial condition) is

$$\boxed{u(x,y) = \frac{3}{2x^2 + \frac{1}{y}} \qquad y \neq 0}$$

<u>Example 3</u> : Suppose now we are given $\boxed{u_x + 3u_y - u = 0}$

We'll solve this two ways : <u>directly</u> and <u>also</u> using the <u>coordinate method</u>

<u>Directly</u>: $u_x + 3u_y = u$ $\quad u \equiv 0$ is a solution

But how about a nontrivial solution? Suppose $\exists$ some $(x_0, y_0) \,/\, u(x_0, y_0) \neq 0$.

Collect as domain of validity all $(x_0, y_0) \,/\, u(x_0, y_0) \neq 0$

Now in the domain of validity the equation can be rewritten as

$$\frac{u_x}{u} + \frac{3u_y}{u} = 1 .$$

constant forcing term

Now let $V(x,y) = \ln|u(x,y)| \Rightarrow \boxed{V_x + 3V_y = 1}$

Now we're reduce to solving an <u>inhomogeneous transport equation with constant coefficients and constant forcing term</u>.

⚠ <u>So</u> let's consider this problem more generally:

$$a\,v_x + b\,v_y = d, \quad d \text{ const.} \quad a, b, d \neq 0$$

BASIC PRINCIPLE: General solution to the INHOMOGENEOUS equation can be obtain as the sum of the general solution to the HOMOGENEOUS equation plus a particular solution of the inhomogeneous equation.

General sol. to HOMOGENEOUS EQUATION	+	A PARTICULAR solution of inhomogeneous EQ.

Now we already know that the general solution to the homogeneous equation $a\tilde{v}_x + b\tilde{v}_y = 0$ has the form $\tilde{v}(x,y) = f(bx - ay)$ for any differentiable function f of one variable. On the other hand, note that if we choose $v_{sp}(x,y) = \frac{d}{a}x$ then v_{sp} is a special solution of $a v_x + b v_y = d$.

Hence any

$$\boxed{v(x,y) = f(bx - ay) + \frac{d}{a}x} \quad (+)$$

<u>will</u> <u>satisfy</u> $\boxed{a v_x + b v_y = d}$

<u>Remark</u>: Note that the function $v_{sp} = \frac{d}{b} y$ is also a special solution of $a v_x + b v_y = d$. So we could have also written the general solutions of $a v_x + b v_y = d$ as

$$\boxed{v(x,y) = f(bx - ay) + \frac{d}{b} y} \quad (\dagger\dagger)$$

for any differentiable function f of one variable. Either general expression is valid since at this point f is generic; so given auxiliary condition determine different f in each case $(\dagger)$ and $(\dagger\dagger)$. One just needs to choose one.

<u>Class homework</u>: Check that both expressions $(\dagger)$ and $(\dagger\dagger)$ for $v(x,y)$ satisfy $a v_x + b v_y = d$.

Back to our particular case (at the bottom of page 8)

$$\boxed{V_x + 3V_y = 1} \qquad (\underline{a = 1 \quad b = 3 \quad d = 1})$$

We now know that the general solution to this inhomogeneous equation can be written as

$$V(x, y) = f(3x - y) + x \qquad \left(\frac{d}{a} = 1\right)$$

Hence the solution u to $u_x + 3u_y - u = 0$ (top of page 8) is

$$u(x, y) = e^{(f(3x-y)+x)}$$

recall
$v = \ln|u|$
$\Rightarrow u = e^v$

$$\Leftrightarrow \boxed{u(x, y) = h(3x - y) \cdot e^x}$$

for any function h of 1 variable

We used that
$e^{(f(3x-y)+x)} = e^{f(3x-y)} \cdot e^x = h(3x-y) \cdot e^x$

<u>Check</u> : We check $u(x,y) = h(3x-y)\cdot e^{x}$ for any (differentiable) function h of 1 variable is indeed a solution of $u_x + 3u_y - u = 0$:

$$u_x = h'(3x-y).3.e^{x} + h(3x-y).e^{x} \quad \left(\begin{array}{c}\text{Chain rule}\\ + \\ \text{Product rule}\end{array}\right)$$

$$u_y = h'(3x-y).(-1).e^{x}$$

$$\therefore u_x + 3u_y - u = \underbrace{\cancel{3e^{x}h'} + \cancel{e^{x}.h}}_{u_x} \underbrace{\cancel{-3h'.e^{x}}}_{+3u_y} \underbrace{\cancel{-h.e^{x}}}_{-u}$$

$$= 0$$

as desired. #